

PHYSICS21-05-2023

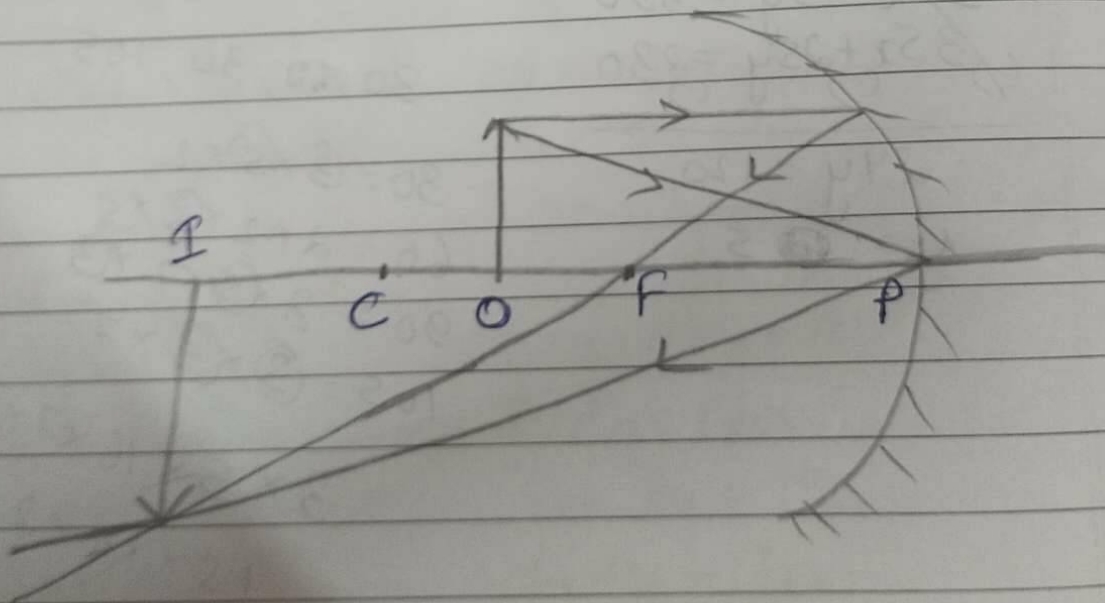
1. Parallel beam of light

2. $f = \frac{R}{2}$

$\Rightarrow f = 25 \text{ cm}$

It will provide convex mirror as its outward bulging surface of the sphere is to be used as mirror.

3.



When the object is placed between C and F the image will be formed beyond C and would be real, enlarged and inverted.

4. height of object = 2cm
 $f = 10\text{cm}$
 $u = 15\text{cm}$

$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$$

$$= \frac{1}{10} = \frac{1}{15} + \frac{1}{v}$$

$$\Rightarrow \frac{1}{v} = \frac{1}{10} - \frac{1}{15}$$

$$\Rightarrow \frac{1}{v} = \frac{3-2}{30} = \frac{1}{30}$$

$$\therefore v = 30\text{cm}$$

Image is formed at a distance of 30cm from the mirror.

$$m = -\frac{v}{u} = -\frac{30}{15} = -2$$

The image of the is diminished and it real and inverted.